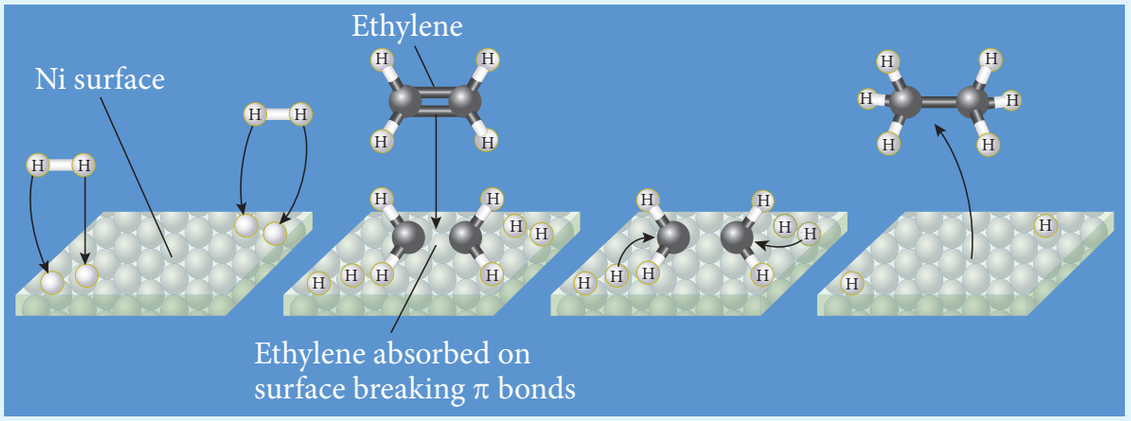


Figure 10.3 Hydrogenation of ethylene in presence of a nickel catalyst.

Active centres

The surface of a catalyst is not smooth. It bears steps, cracks and corners. Hence the atoms on such locations of the surface are co-ordinatively unsaturated. So, they have much residual force of attraction. Such sites are called active centres. So, the surface carries high surface free energy.

The presence of such active centres increases the rate of reaction by adsorbing and activating the reactants.

The adsorption theory explains the following

i. Increase in the surface area of metals and metal oxides by reducing the particle size increases acting of the catalyst and hence the rate of the reaction.

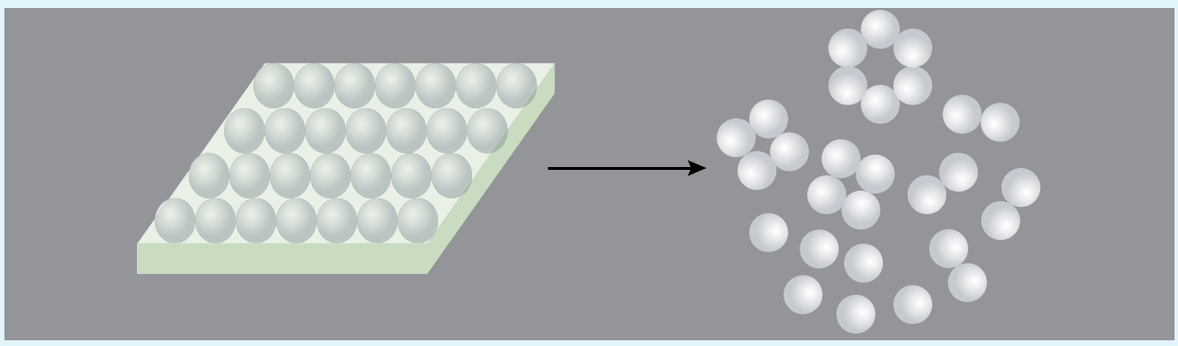


Figure 10.4 Finely divided catalyst is more effective due to increase in the number of active centres.

ii. The action of catalytic poison occurs when the poison blocks the active centres of the catalyst.

iii. A promoter or activator increases the number of active centres on the surfaces.

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